

CS51012 GPU Programming and real-time rendering

Individual Action Scene: 2025/2026

Student: [Tim Clarke]

Grade: Part 1 [A2], Part 2 [A2], Final grade [A2]

PART 1

PART1: Core requirements:

Requirement	Excellent (A)	Good (B)	Acceptable ©	Problematic	Not included
Two or more 3D objects with moving parts	☒	☐	☐	☐	☐
Lighting	☒	☐	☐	☐	☐
Camera controls	☒	☐	☐	☐	☐
User controls	☒	☐	☐	☐	☐
Realism of scene / quality of graphics	☒	☐	☐	☐	☐
Code style and structure	☒	☐	☐	☐	☐
Submission(runnable exe, source code, cleaned)	☒	☐	☐	☐	☐

PART1: Suggestions for Improvements

No suggestions for this Part, you have submitted an assignment that is significantly above and beyond what was requested. Your models and lighting in particular are excellent.

PART 1: Overall comments

Excellent assignment, I have no criticisms. You met the action brief, lighting and all other aspects to a high standard. I also particularly liked your documentation (I do like a good uml diagram!) which was not specifically requested but was very helpful, and your clarity of how you used AI tools was also exemplary.

For Part 2, I would recommend placing your mobile in either a room or a fantasy scene. This would give you the opportunity to demonstrate many other potential techniques such as importing mesh models, normal mapping, particle animations (they seem

particularly appropriate for your ethereal-style scene). You have already put a lot of effort into your lighting I also think that placing your mobile in the context of a wider scene would give you an opportunity to both further demonstrate and “show” off different lighting effects (e.g. a room lamp plus the mobile lights) and material properties.

PART 2 Core requirements:

Requirement	Excellent (A)	Good (B)	Acceptable (C)	Problematic	Not included
Background	☒	☐	☐	☐	☐
Use of external objects	☒	☐	☐	☐	☐
Texturing	☒	☐	☐	☐	☐
Advanced Concepts	☒	☐	☐	☐	☐
User Camera controls (appropriate for scene)	☒	☐	☐	☐	☐
Animation & Lighting (appropriate for scene)	☒	☐	☐	☐	☐
Realism of scene / quality of graphics	☒	☐	☐	☐	☐
Code style and structure	☒	☐	☐	☐	☐
Submission (runnable exe, source code, video, cleaned)	☒	☐	☐	☐	☐
Report	☒	☐	☐	☐	☐

PART 2: Additional features:

A lot going on, normal mapping, textured skybox, particle animations on the orbs

PART 2: Suggestions for Improvements

Nothing to suggest, you did more than enough for this assignment.

PART 2: Overall comments

Great assignment, I particularly liked the way that all the features you implemented added to the scene in a coherent way. Your code structure is excellent, and the video shows off your features nicely – great effort.